Lesson Plan

Teacher Name: Ms. Lubna Kakkengal and Mr. Dilan Devendra	**Class/Section:** Grade 6
Subject: TCD: Photon AI Mission	**Date: Week 13 (23 Nov to 27 Nov)**
Class Description: Grade 6 TCD class, Photon AI Mission Session 2 of 2 (2 hours). Students add selection (if-then sensor conditions) and repetition (loops) so Photon can react to obstacles, then complete a mission challenge with a troubleshooting log and peer debugging.	
TA/LSA Role (if applicable):	

Differentiated Learning Groups (Data)			
Desert Fox	**Camel**	**Oryx**	**Falcon**

Lesson Overview		
Lesson Topic	Selection and Loops Mission	
Standard(s)	Technology: selection and repetition in algorithms, debugging; Math: 6.EE.B (reasoning with variables); ELA: SL.6.1 (collaborative discussion)	
Learning Objective(s) (WALT): We are learning to use if-then conditions and loops so a robot can make decisions and repeat actions.		**Success Criteria (WILF):** I can explain selection and repetition. I can program Photon to react when its sensor detects an obstacle. I can use a loop so my program keeps checking the sensor. I can log at least two errors and fix them, including one found by a peer.
Literacy Strategies	Condition frame: If Photon senses ..., then it ..., else it ...; troubleshooting log with sentence starters.	
Assessment	Mission success; troubleshooting log with two errors; peer debugging notes; live quiz on the TCD website (counts towards the Photon AI Mission certificate).	
Targeted Vocabulary	selection, condition, if-then-else, repetition, loop, sensor, obstacle, bug, optimise	
Skills	Logical reasoning, debugging, collaboration, persistence	
National Identity	Link to driverless vehicles being tested in Dubai and Abu Dhabi,	

Integration	which use sensors and if-then decisions to avoid obstacles safely.
AI/Digital Learning	Photon robots and tablets with the Photon coding app; hook video; live quiz on the TCD website. Presentation: Grade6_Week13_Presentation.pptx. Worksheet: Grade6_Week13_Worksheet.

Lesson Structure	
Starter (Hook) Engage and Ask (15 minutes)	Warm-up: If This, Then That (slide 4). Show the hook video and ask the hook question (slide 2). Share the learning goals (slide 3) and key words (slide 5).
I Do (25 minutes)	Teach selection with flowchart diamonds (slide 6). Teach loops and why the condition goes inside the loop (slide 7). Model troubleshooting with a deliberate bug (slide 8).
We Do (15 minutes)	Class draws one obstacle flowchart together. Pairs code a single obstacle check and test it (slide 10).
You Do (50 minutes)	Pairs complete the obstacle mission and keep the troubleshooting log (slide 12). Peer debugging swap for 10 minutes (slide 9). Challenge levels on the board (slide 13). Live quiz on tablets (slide 14).
Differentiated Instruction: **Beginning / On / Above**	**Beginning: Use the flowchart template and code one obstacle check inside a loop.** **On: Complete the full obstacle mission with a working if-then-else inside a loop.** **Above: Add a second condition (for example a sound or light reaction) and explain how your optimisation made the program better.**
Closure/ Reflection (15 minutes)	Pairs share the best bug they found and how they fixed it. Exit ticket (slide 15). Robots back to charging.

Inclusive Planning
Students receiving additional inclusive support (IEP or ILPs) or Gifted and Talented: EAL: condition sentence frame and log sentence starters. SEN: flowchart template and single-obstacle version. G&T: second condition and an explanation of the optimisation; lead peer debugging.
Inclusion / EAL Support

☑ Visual Supports
☐ Chunked Instructions
☐ Reduced Language Load
☑ Model / Worked Example
☐ Pre-teach Vocabulary
☐ TA/LSA Support
☐ Extra Processing Time
☐ Text Read by TA/LSA
☑ Flexible Grouping
☐ Assistive Technology
☐ Movement Break
☑ Scaffold / Sentence Starters
☐ Other:

G&T / Extension

☑ Extension / Challenge Task
☑ Higher-order Questioning
☐ Depth & Complexity Task
☐ Independent Inquiry
☑ Leadership / Peer Coaching
☐ Real-World Application
☐ Curriculum Compacting
☐ Above-grade-level Texts
☐ Student-led / Passion Project
☐ Open-Ended Task
☐ Other:

Assessment Adaptations

☑ Oral Response
☐ Reduced Questions
☑ Scaffolded Success Criteria
☐ Extra Time
☐ Alternative Outcome Format
☑ Teacher Conferencing/Live Feedback
☐ Quiet / Separate Space
☐ Other: